MCV4U1 – V2
Chang, Moshtagh, Vincent

Name: Solutions **Alpha #:** ______

UNIT 1 ASSESSMENT OF LEARNING: RATES OF CHANGE AND LIMITS – DAY 1

K & U	Thinking	Comm.
/22	/5	/2

Instructions: You MUST use concepts covered in this unit/course. The Slope of the Tangent or the Derivative MUST be done using First Principles. Show all steps for full marks. The use of cellphones, audio or video recording devices, digital music players or email or text-messaging devices during the assessment is prohibited.

KNOWLEDGE & UNDERSTANDING - [22 Marks]

Multiple Choice: Write the CAPITAL LETTER corresponding to the correct answer on the line provided.
[1 Mark Each – 5 Marks Total]

1. Which of the following is/are **false**? B

   I. If $f(x)$ has a cusp at $x = a$, then no unique tangent exits at $x = a$.

   II. If $f(x)$ has a vertical tangent at $x = a$, then $\left.\frac{dy}{dx}\right|_{x=a} = 0$.

   III. If $f(x) = \frac{\sqrt{7-x}}{3}$, then $\lim_{x \to 7^+} f(x) = 0$.

   IV. $\lim_{x \to 2} \frac{x^2+5x+6}{2x^2+7x+3} = \frac{\lim_{x \to 2} x^2+5x+6}{\lim_{x \to 2} 2x^2+7x+3}$.

   A. I, II and IV only. B. II and III only. C. III and IV only. D. All are false.

2. Evaluate $\lim_{x \to \sqrt{5}} \frac{x^2-5}{x-\sqrt{5}}$, if it exists. A

   A. $2\sqrt{5}$ B. $-2\sqrt{5}$ C. 0 D. Does Not Exist

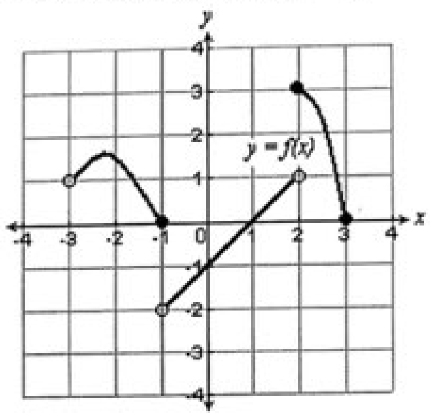
3. Which of the following is/are **true**? C

   I. If $f(x)$ has an infinite discontinuity at $x = a$, then there must be a hole at $x = a$.

   II. If $\lim_{x \to a} f(x)$ does not exist, then there can be a jump discontinuity at $x = a$.

   III. If $f(x)$ is continuous at $x = a$, then $f(x)$ must be differentiable at $x = a$.

   IV. The graph below has a removable discontinuity at $x = -1$.



   A. I and III only. B. II, III and IV only. C. II only. D. II and IV only.

4. $\lim_{x \to -\infty} \frac{1-7x^3}{(3x^2+8)(1-6x)}$ equals C

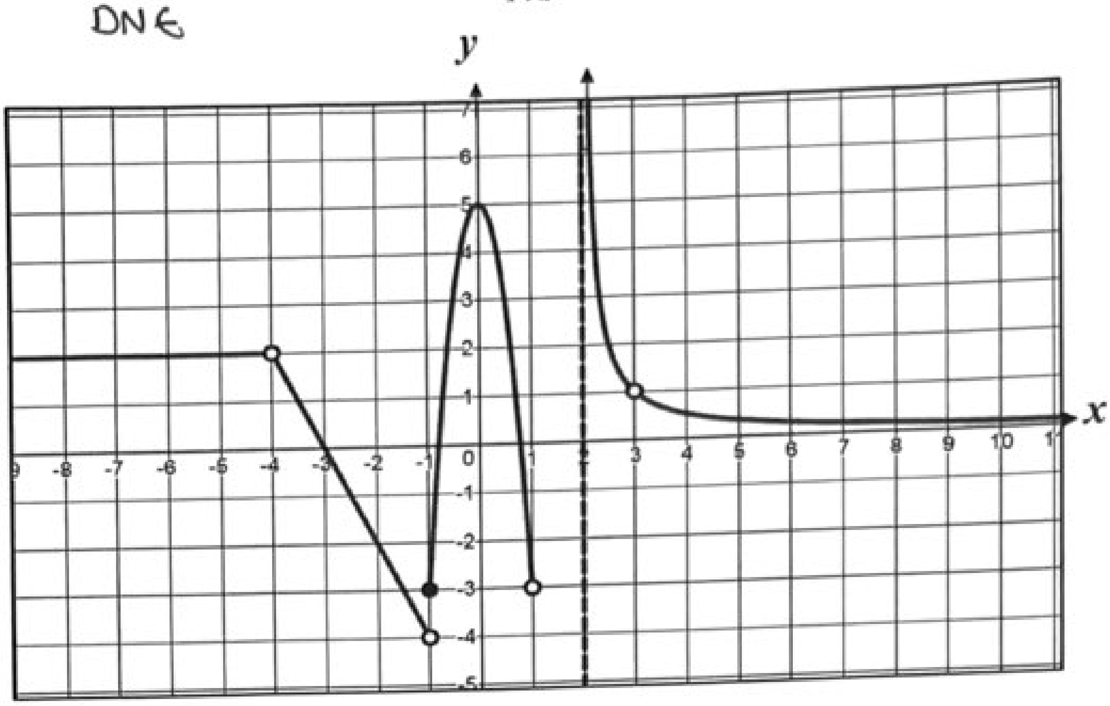
   A. $-\frac{7}{3}$ B. $-\infty$ C. $\frac{7}{18}$ D. $\frac{1}{3}$

5. If $\lim_{x \to a^+} f(x) = -\infty$, the graph of $f(x)$ has A

   A. a vertical asymptote at $x = a$. B. a removable discontinuity at $x = a$.

6. Given the function $y = f(x)$ below, determine the following: **[0.5 Marks Each. 2 Marks Total]**

a. $\lim_{x\to 2^+} f(x) =$ ∞ or DNE b. $f(-1) =$ -3 c. $\lim_{x\to\infty} f(x) =$ 0 d. $\lim_{x\to -4} f(x) =$ 2



7. Evaluate each the following limits, if they exist. **[12 Marks]**

a. $\lim_{x\to\sqrt{8}} \frac{x^2-8}{4-\sqrt{8+x^2}}$ **[3]**

$= \lim_{x\to\sqrt{8}} \frac{(x^2-8)(4+\sqrt{8+x^2})}{(4-\sqrt{8+x^2})(4+\sqrt{8+x^2})}$

$= \lim_{x\to\sqrt{8}} \frac{(x^2-8)(4+\sqrt{8+x^2})}{16-8-x^2}$

$= \lim_{x\to\sqrt{8}} -\frac{\cancel{(x^2-8)}(4+\sqrt{8+x^2})}{\cancel{8-x^2}}$

$= \lim_{x\to\sqrt{8}} -(4+\sqrt{8+x^2})$

$= -8$

b. $\lim_{x\to 1} \frac{x^3-1}{x^3+7x^2-x-7}$ **[3]**

$= \lim_{x\to 1} \frac{(x-1)(x^2+x+1)}{x^2(x+7)-(x+7)}$

$= \lim_{x\to 1} \frac{\cancel{(x-1)}(x^2+x+1)}{(x+7)\cancel{(x-1)}(x+1)}$

$= \lim_{x\to 1} \frac{x^2+x+1}{(x+7)(x+1)}$

$= \frac{3}{16}$

c. $\lim_{x\to 1}\left[\left(\frac{1}{x-1}\right)\left(1-\frac{36}{(x+5)^2}\right)\right]$ **[3]**

$= \lim_{x\to 1}\left[\left(\frac{1}{x-1}\right)\left(\frac{(x+5)^2-36}{(x+5)^2}\right)\right]$

$= \lim_{x\to 1}\left[\left(\frac{1}{x-1}\right)\left(\frac{x^2+10x+25-36}{(x+5)^2}\right)\right]$

$= \lim_{x\to 1}\left[\left(\frac{1}{x-1}\right)\left(\frac{x^2+10x-11}{(x+5)^2}\right)\right]$

$= \lim_{x\to 1}\left[\left(\frac{1}{\cancel{x-1}}\right)\left(\frac{\cancel{(x-1)}(x+11)}{(x+5)^2}\right)\right]$

$= \lim_{x\to 1} \frac{x+11}{(x+5)^2}$

d. $\lim_{x\to 21} \frac{\sqrt[3]{x+6}-3}{x-21}$ **[3]**

Let $t = \sqrt[3]{x+6}$ As $x \to 21$

$t^3 - 6 = x$ $t \to 3$

$\lim_{t\to 3} \frac{t-3}{t^3-27}$

$= \lim_{t\to 3} \frac{\cancel{t-3}}{\cancel{(t-3)}(t^2+3t+9)}$

$= \lim_{t\to 3} \frac{1}{t^2+3t+9}$

$= \frac{1}{27}$

8. Given $\lim_{x\to 1} f(x)=4$ and $\lim_{x\to 1} g(x)=8$, use the **properties of limits** to evaluate $\lim_{x\to 1}\dfrac{\sqrt{f(x)}+\sqrt{g(x)+1}}{3-[f(x)]^2}$.
[3 Marks]

$$=\frac{\sqrt{\lim_{x\to 1} f(x)}+\sqrt{\lim_{x\to 1} g(x)+\lim_{x\to 1} 1}}{\lim_{x\to 1} 3-\left[\lim_{x\to 1} f(x)\right]^2}$$

$$=\frac{2+3}{3-16}=\frac{-5}{13}$$

THINKING - [5 Marks]

1. Determine all the equations (point-slope form) of the tangents to the graph of $f(x)=2x-8x^3$ where $f(x)$ intersects the $x-$axis. **[5 Marks]**

x-intercepts

$f(x)=2x(1-4x^2) \Rightarrow$ x-ints @ $x=0, \pm\frac{1}{2}$

points are $(0,0), (\frac{1}{2},0), (\frac{-1}{2},0)$

$$m_t=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$=\lim_{h\to 0}\frac{2(x+h)-8(x+h)^3-(2x-8x^3)}{h}$$

$$=\lim_{h\to 0}\frac{2x+2h-8(x^3+3x^2h+3xh^2+h^3)-2x+8x^3}{h}$$

$$=\lim_{h\to 0}\frac{\cancel{2x}+2h-\cancel{8x^3}-24x^2h-24xh^2-8h^3-\cancel{2x}+\cancel{8x^3}}{h}$$

$$=\lim_{h\to 0}\frac{\cancel{h}(2-24x^2-24xh-8h^2)}{\cancel{h}}$$

$$=\lim_{h\to 0} 2-24x^2-24xh-8h^2$$

$$=2-24x^2$$

$\therefore m_t|_{x=0}=2$

$y-y_1=m(x-x_1)$

$y-0=2(x-0)$

$y=2x$

$m_t|_{x=\frac{1}{2}}=-4$

$y-y_1=m(x-x_1)$

$y=-4(x-\frac{1}{2})$

$m_t|_{x=-\frac{1}{2}}=-4$

$y-y_1=m(x-x_1)$

$y=-4(x+\frac{1}{2})$

*** 2 Marks will be awarded in the Communication Category for proper mathematical form. ***

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UNIT 1 ASSESSMENT OF LEARNING: RATES OF CHANGE AND LIMITS – DAY 2

Application	Thinking	Comm.
/20	/5	/5

Instructions: You MUST use concepts covered in this unit/course. The Slope of the Tangent or the Derivative MUST be done using First Principles. Show all steps for full marks. The use of cellphones, audio or video recording devices, digital music players or email or text-messaging devices during the assessment is prohibited.

APPLICATION - [20 Marks]

1. Katie kicks a soccer ball off the top of a building that follows the path given by the function $H(t) = 121 - 4.9t^2$, where $h(t)$ is the height, in metres, above the ground after t seconds.

   a. Determine the average speed of the soccer ball between $t = 0$ seconds and $t = 1$ second. **[1 Mark]**

$$AROC = \frac{H(1) - H(0)}{1-0}$$

$$AROC = 121 - 4.9 - 121$$

$$AROC = -4.9 \text{ m/s}$$

   b. Determine the speed of the soccer ball at $t = 2$ seconds. **[3 Marks]**

$$\lim_{h\to 0} \frac{H(t+h) - H(t)}{h}$$

$$= \lim_{h\to 0} \frac{121 - 4.9(t+h)^2 - (121 - 4.9t^2)}{h}$$

$$= \lim_{h\to 0} \frac{-4.9(t^2 + 2th + h^2) + 4.9t^2}{h}$$

$$= \lim_{h\to 0} \frac{\cancel{-4.9t^2} - 9.8th - 4.9h^2 \cancel{+ 4.9t^2}}{h}$$

$$= \lim_{h\to 0} \frac{-\cancel{h}(9.8t + 4.9h)}{\cancel{h}} = -9.8t \Rightarrow IROC = -19.6 \text{ m/s}$$

2. Determine the point(s) on the curve $f(x) = \dfrac{x+4}{2x+9}$ where the tangent line is perpendicular to the line $x + y - 12 = 0$. **[5 Marks]**

$$y = -x + 12 \Rightarrow m = -1 \Rightarrow m_t = 1$$ (negative reciprocal)

$$\lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{\frac{x+h+4}{2(x+h)+9} - \frac{x+4}{2x+9}}{h}$$

$$= \lim_{h\to 0} \frac{(x+h+4)(2x+9) - (2(x+h)+9)(x+4)}{h\cdot(2(x+h)+9)(2x+9)}$$

$$= \lim_{h\to 0} \frac{\cancel{2x^2} + \cancel{2xh} + \cancel{8x} + \cancel{9x} + 9h + \cancel{36} - (\cancel{2x^2} + \cancel{8x} + \cancel{2xh} + 8h + \cancel{9x} + \cancel{36})}{h(2(x+h)+9)(2x+9)}$$

$$= \lim_{h\to 0} \frac{\cancel{h}}{\cancel{h}(2(x+h)+9)(2x+9)}$$

$$= \frac{1}{(2x+9)^2}$$

$$\therefore \frac{1}{(2x+9)^2} = 1$$

$$2x + 9 = \pm 1$$

$x = -5$ $(-5, 1)$

$x = -4$ $(-4, 0)$

3. A curve is defined by $f(x)=a\sqrt{x}$, where a is a constant. Given that $m_t=5$ at $x=4$ determine the value of a. **[3 Marks]**

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$=\lim_{h\to 0}\frac{a\sqrt{x+h}-a\sqrt{x}}{h}$$

$$=a\lim_{h\to 0}\frac{(\sqrt{x+h}-\sqrt{x})(\sqrt{x-h}+\sqrt{x})}{h(\sqrt{x-h}+\sqrt{x}}$$

$$=a\lim_{h\to 0}\frac{x+h-x}{h(\sqrt{x-h}+\sqrt{x})}$$

$$=\frac{a}{2\sqrt{x}}$$

$$\therefore \frac{a}{2\sqrt{4}}=5$$

$$a=20$$

4. Determine the values p and q such that the function $f(x)=\begin{cases} x^2+px+4 & , x\in(-\infty,3) \\ p+q & , x=3 \\ -\sqrt{x^2+7}+2q & , x\in(3,\infty) \end{cases}$

is continuous on the interval $(-\infty,\infty)$. **[4 Marks]**

① $f(3)=p+q$ (defined)

② $\lim_{x\to 3^-} f(x)$ | $\lim_{x\to 3^+} f(x)$

$=\lim_{x\to 3^-} x^2+px+4$ | $=\lim_{x\to 3^+} -\sqrt{x^2+7}+2q$

$=13+3p$ | $=-4+2q \Rightarrow 13+3p=-4+2q$

③ $p+q=13+3p$ | $p+q=-4+2q$

$\boxed{2p-q=-13}$ Ⓐ | $\boxed{p-q=4}$ Ⓑ

Ⓐ - Ⓑ

$2p-q=-13$

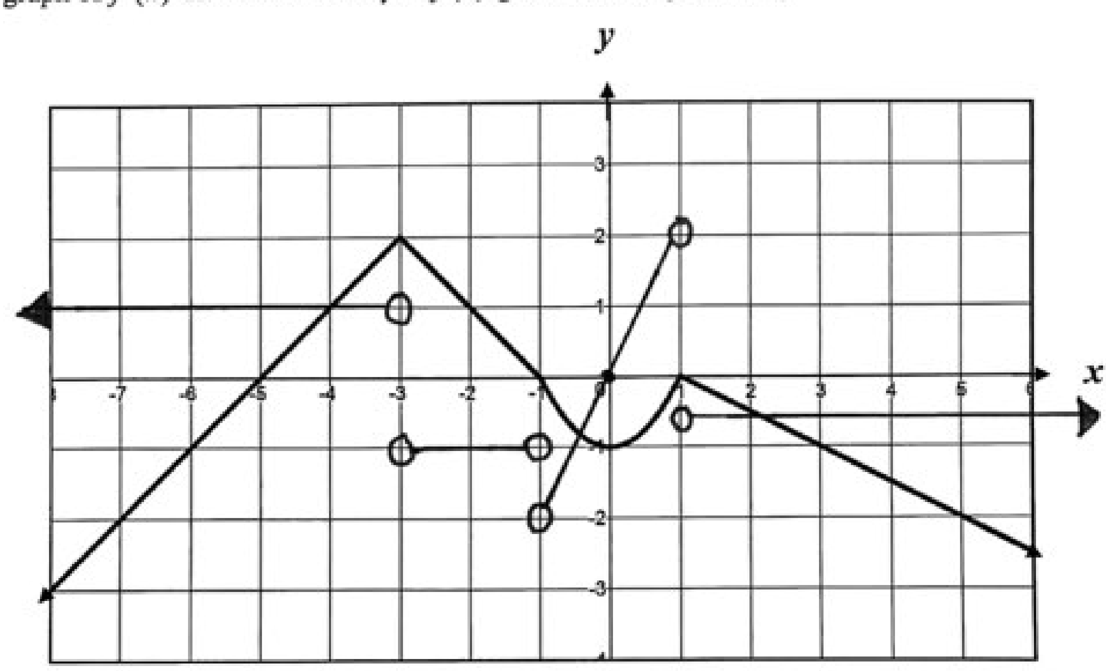
$-(p-q=-4)$

$p=-9$

sub $p=-9$ into Ⓑ

$\Rightarrow q=-5$

5. Sketch the graph of $f'(x)$ for the function $y=f(x)$ given below. **[4 Marks]**



THINKING - [5 Marks]

1. Evaluate $\lim_{x \to 2} \frac{x^2 - 4}{x^4 + x^3 - 3x^2 + kx - 2}$ given that the limit exists and k is a constant. **[5 Marks]**

$$= \lim_{x \to 2} \frac{(x-2)(x+2)}{x^4 + x^3 - 3x^2 + kx - 2} = \frac{0}{0} \Big\} \text{since } \lim_{x\to 2} f(x) \text{ exists}$$

$$\Rightarrow (2)^4 + (2)^3 - 3(2)^2 + 2k - 2 = 0$$

$$16 + 8 - 12 + 2k - 2 = 0$$

$$k = -5$$

$$\therefore \quad x - 2 \,\overline{\big)\, x^4 + x^3 - 3x^2 - 5x - 2}$$

Quotient: $x^3 + 3x^2 + 3x + 1$

$$-(x^4 - 2x^3)$$
$$3x^3 - 3x^2 - 5x - 2$$
$$-(3x^3 - 6x^2)$$
$$3x^2 - 5x - 2$$
$$-(3x^2 - 6x$$
$$x - 2$$
$$-(x-2)$$
$$0R.$$

$$\lim_{x \to 2} \frac{\cancel{(x-2)}(x+2)}{\cancel{(x-2)}(x^3 + 3x^2 + 3x + 1)}$$

$$= \lim_{x \to 2} \frac{x+2}{x^3 + 3x^2 + 3x + 1}$$

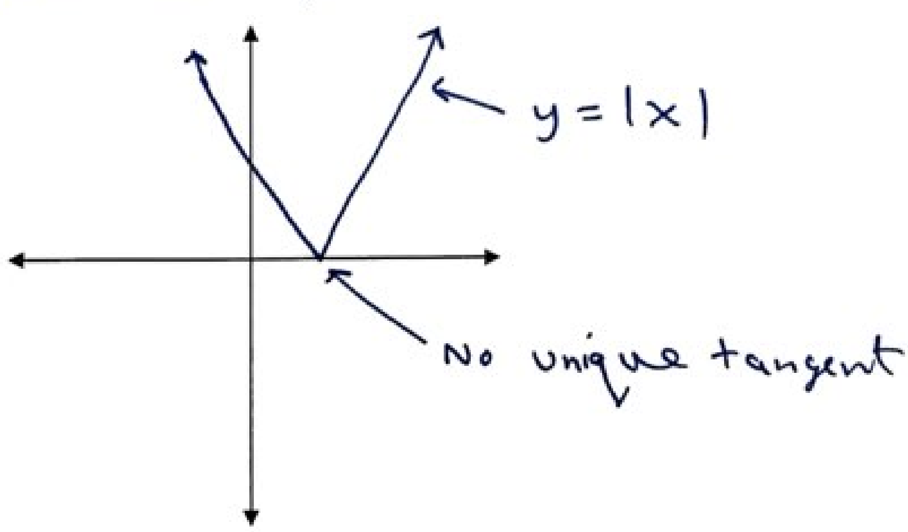
$$= \frac{4}{27}$$

COMMUNICATION - [5 Marks]

1. Explain whether or not it is possible for a continuous function to be non-differentiable at certain points. Use a graphical illustration to support your explanation. **[3 Marks]**

* Yes, it is possible.
* It can occur at a sharp corner or a cusp where no unique tangent exists or at a vertical tangent where the tangent line slope is undefined.

* Answers may Vary!



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